

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

JetBrains annual survey of 23,262 developers worldwide

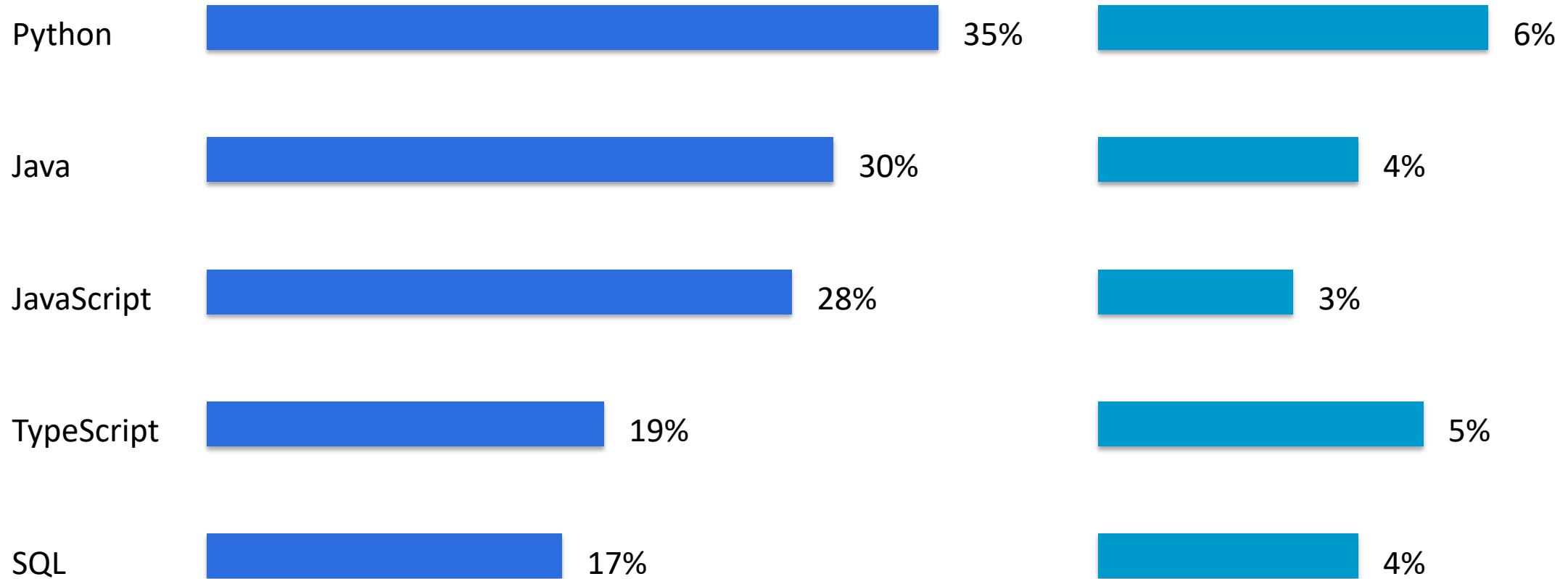
- Most comprehensive 2024 snapshot of languages, tools, AI attitudes, and career trends.
- Focus: how usage is changing, what developers prefer, and strategic actions for 2025.
- Audience: Engineering leaders, CTOs, and developer experience teams.

Python Closes the Gap: 8 Years of Language Usage Shifts

Language	2017	2020	2022	2024	Trend
JavaScript	65%	70%	65%	61%	Declining from peak
Python	32%	55%	53%	57%	Steady growth
HTML/CSS	60%	61%	54%	51%	Gradual decline
SQL	42%	56%	49%	48%	Plateaued
Java	47%	54%	48%	46%	Slow decline
TypeScript	12%	28%	34%	37%	Strong growth
Go	8%	19%	19%	18%	Plateaued after growth
Rust	--	7%	9%	11%	Steady growth
PHP	30%	27%	20%	17%	Significant decline
Kotlin	2%	17%	16%	14%	Plateaued

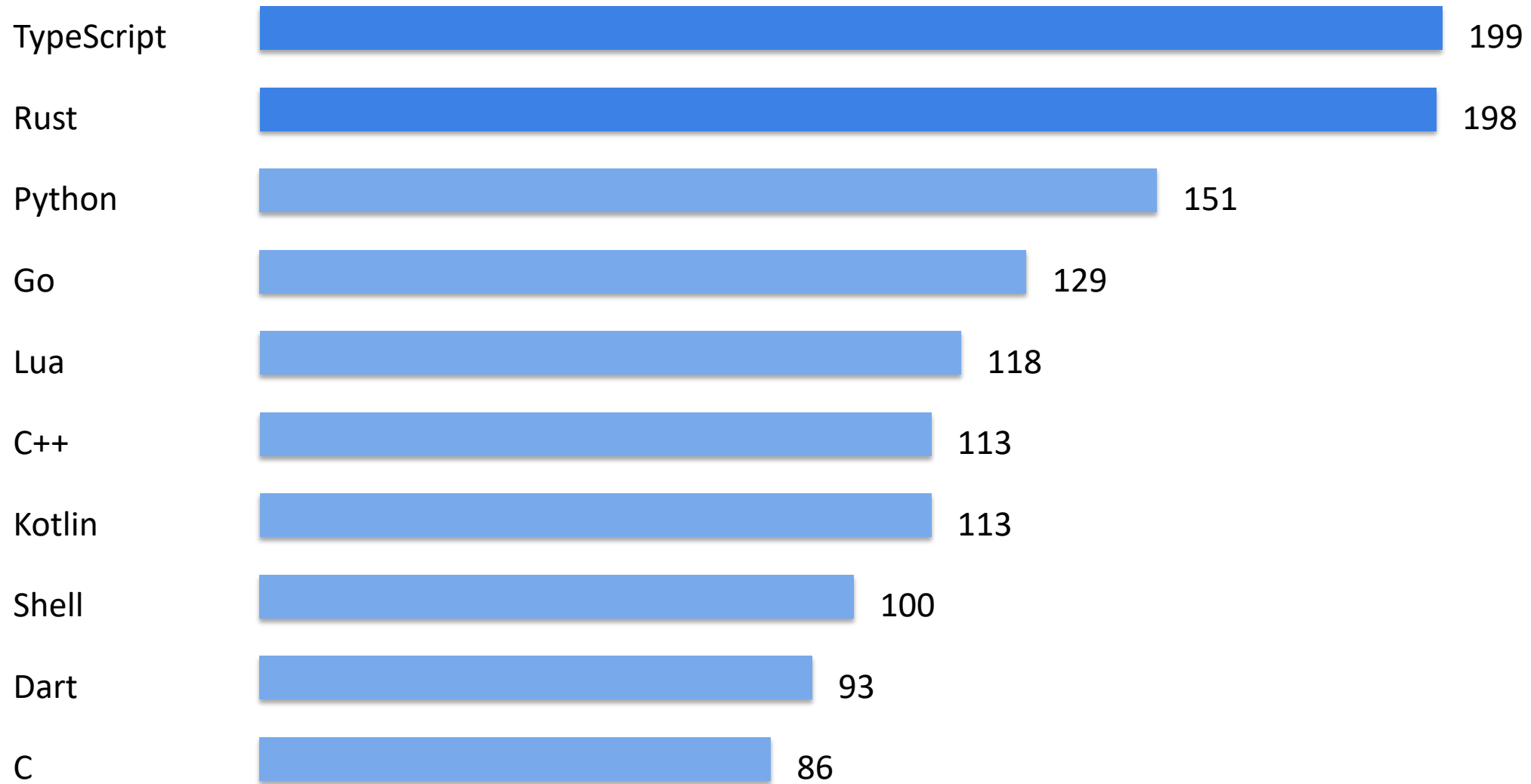
Key signal: JavaScript remains largest by total usage (61%), while Python rapidly closes the gap at 57% and TypeScript triples vs 2017.

Python Is Now the #1 Primary Language at 35%



Blue = Primary language share | Cyan = Planning to adopt

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Leading Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

What This Means for Your Technology Strategy in 2025

- Invest in Python for AI/ML and data teams — clear domain leader (31% AI/ML, 28% data science).
- Prioritize TypeScript for frontend/full-stack roadmaps — strong usage growth and highest satisfaction.
- Build Rust and Go capabilities for infrastructure and systems where reliability and performance matter.
- Plan workforce risk for declining ecosystems: PHP (30%→17% since 2017) and Ruby (10%→4%).
- Use satisfaction signals (TypeScript 199, Rust 198) to guide tooling and retention strategy.